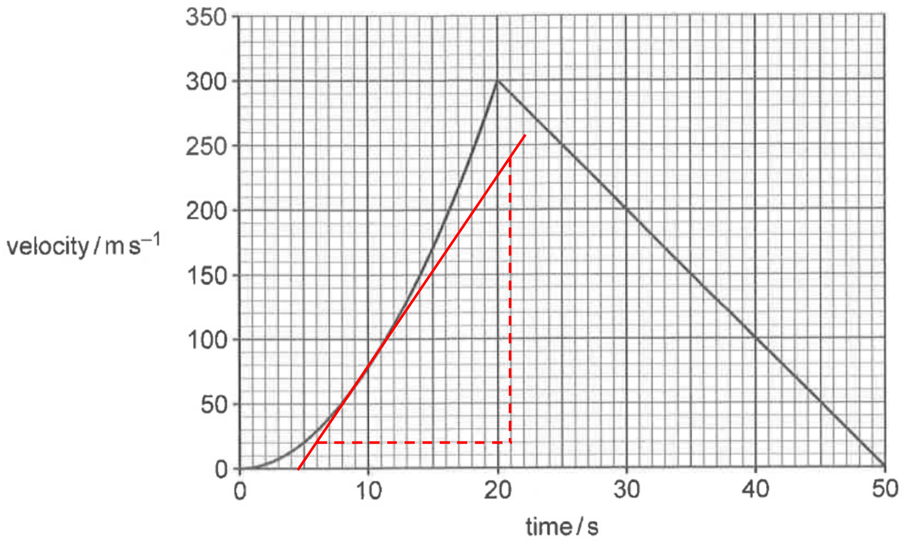
No.	Solution
1(a)	 **Fig. 1.1** Acceleration $= \dfrac{240-20}{21-6} = 14.67 \text{ m s}^{-2} \approx 15 \text{ m s}^{-2}$
1(b)(i)	The rocket reached maximum velocity / had stopped accelerating at t = 20 s. The engines had switched off / the fuel supply had been exhausted.
1(b)(ii)	Acceleration reaches a maximum value just before t = 20 s. The gradient of the graph is greatest at this time.
1(b)(iii)	After t = 20 s, deceleration $= \dfrac{300-0}{30} = 10 \text{ m s}^{-2}$ The rocket undergoes constant deceleration of 10 m s^{-2}. It continues to travel upwards at a decreasing speed, until it reaches maximum height at t = 50 s.
1(c)	The maximum height reached is determined by the area between the line and the time axis from time = 0 to time = 50 s.
2(a)	510